# Humanized CD40 Mouse

**Strain Name: C57BL/6-*Cd40* $^{em1(hCD40)Smoc}$ Strain Background: C57BL/6 Cat. No. : IT-HU-00076**

CD40 is a member of the tumor necrosis factor (TNF) receptor family and is also known as TNFRSF5. CD40 is expressed on antigen presenting cells (APC) such as B cells, dendritic cells (DC), and monocytes as well as many non-immune cells and various types of cancer cells. Upon binding to its ligand CD40L expressed on the antigen presenting cells (APCs), the CD40-CD40L complex plays a vital role in the helper T cells function to prime $CD8^+$ T cells. It has been demonstrated *in vitro* that agonistic CD40 mAb can improve the ability of APCs to cross-prime naive T cells to tumor antigens.

## Construction strategy

The humanized CD40 mice were developed on the C57BL/6 background. The coding sequence for the extracellular domain of the mouse endogenous Cd40 was completely replaced with the human-derived sequence, resulting in the expression of a humanized, chimeric CD40 gene.

## Validation data

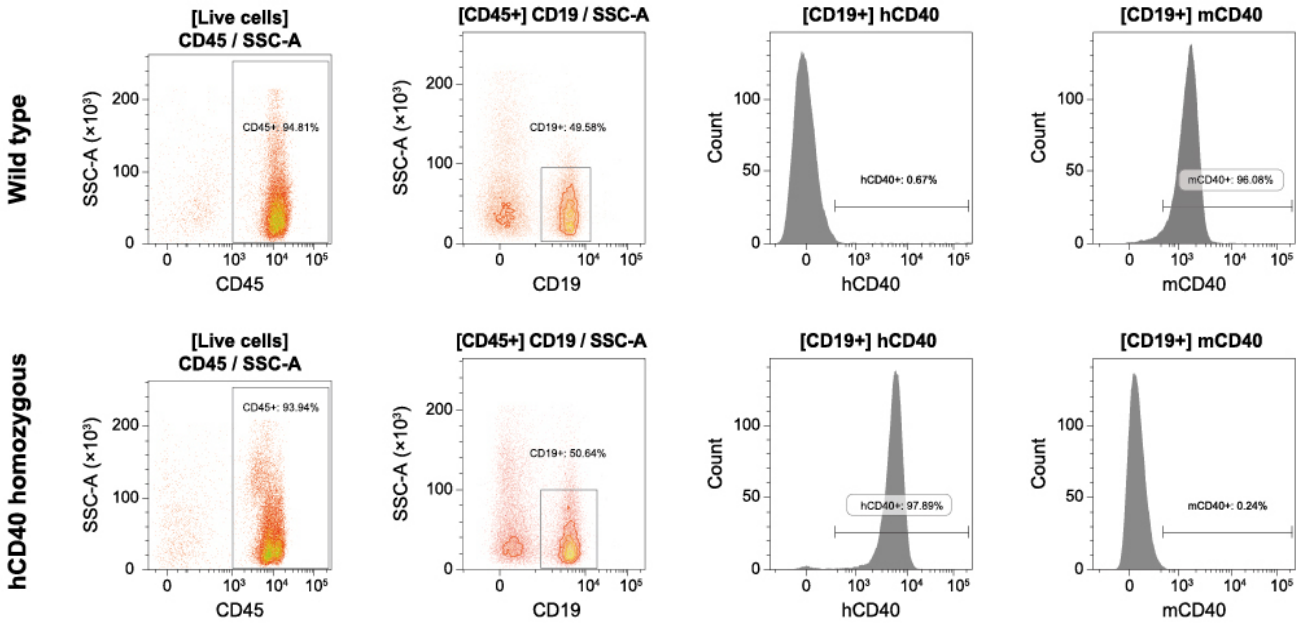



Figure 1. Detection of CD40 expression in peripheral blood cells of humanized CD40 mice. The FACS results of peripheral blood cells collected from homozygous humanized CD40 mice and wild-type mice showed that the active expression of humanized CD40 was detected in CD19 positive cells collected from homozygous humanized CD40 mice, and its expression level was similar to that of murine Cd40 expression in wild-type mice. (Completed in collaboration with CrownBio).

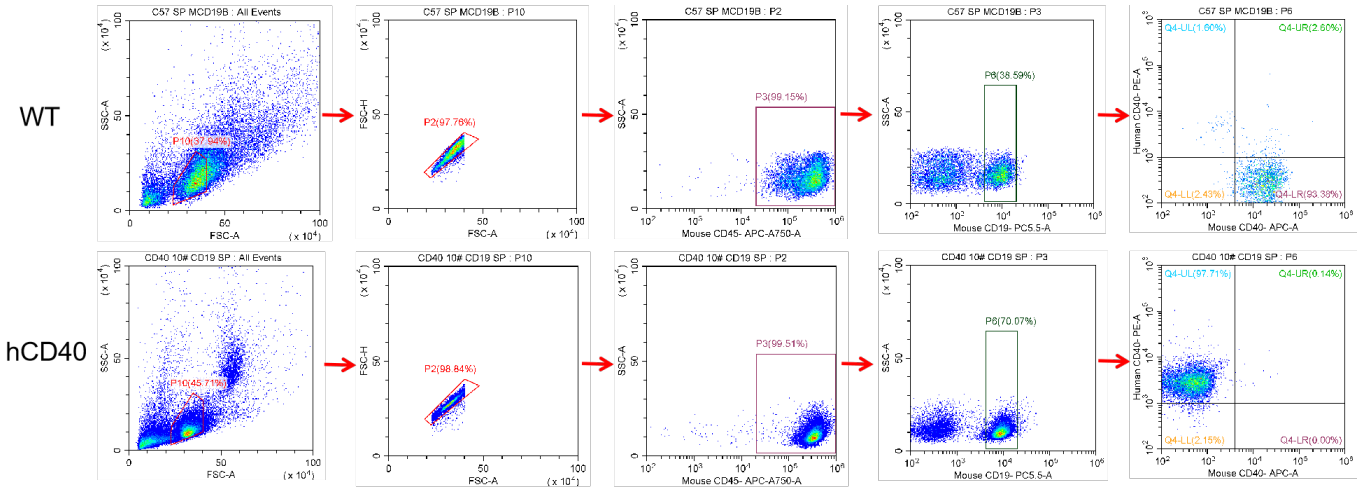
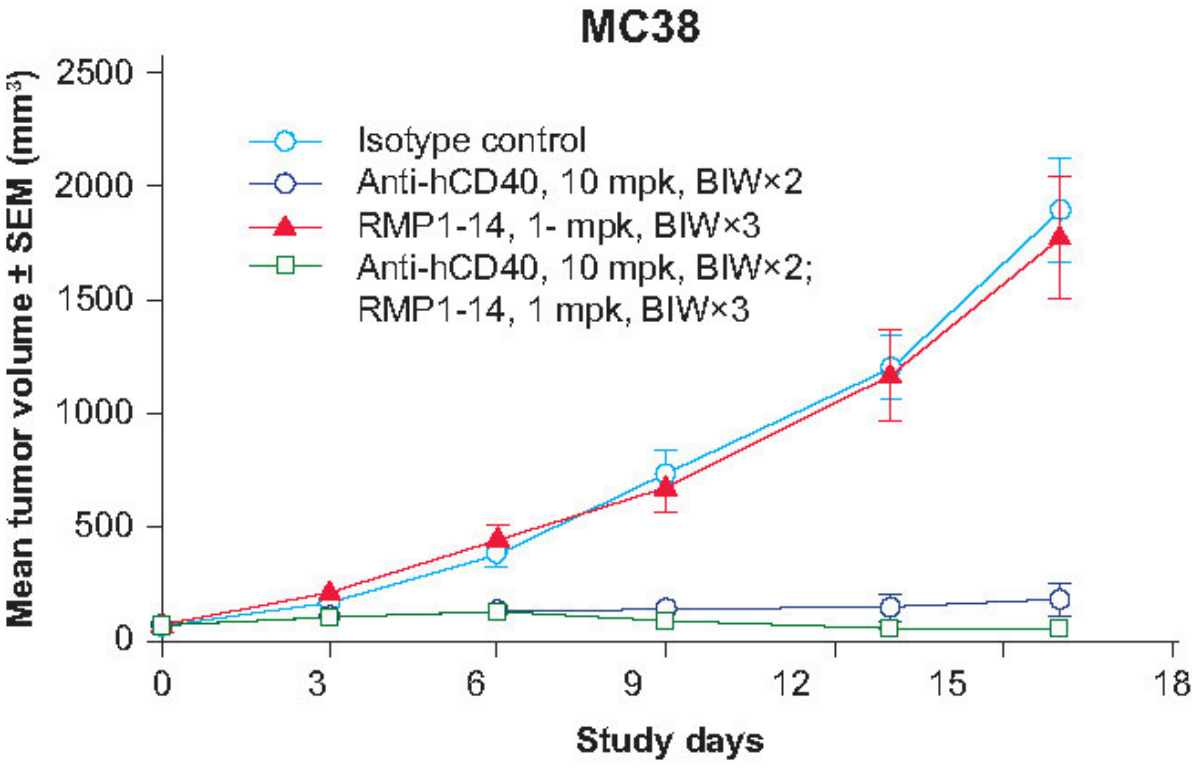



Figure 2. Analysis of human CD40 expression in the spleen by FACS. The homozygous KI animal expresses human CD40 on the CD19+ B cells.

## Application

### Case study 1： In vivo validation in a MC38 tumor-bearing model of humanized CTLA4 mouse



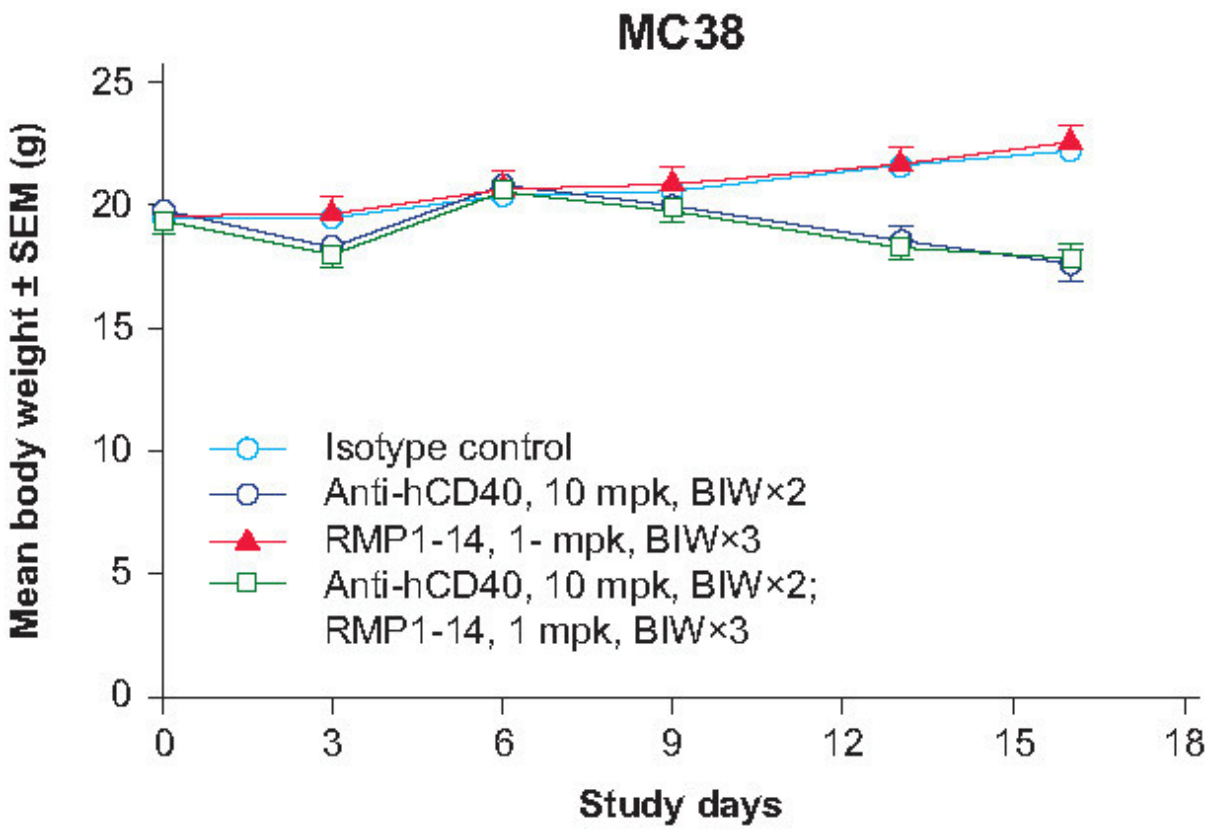
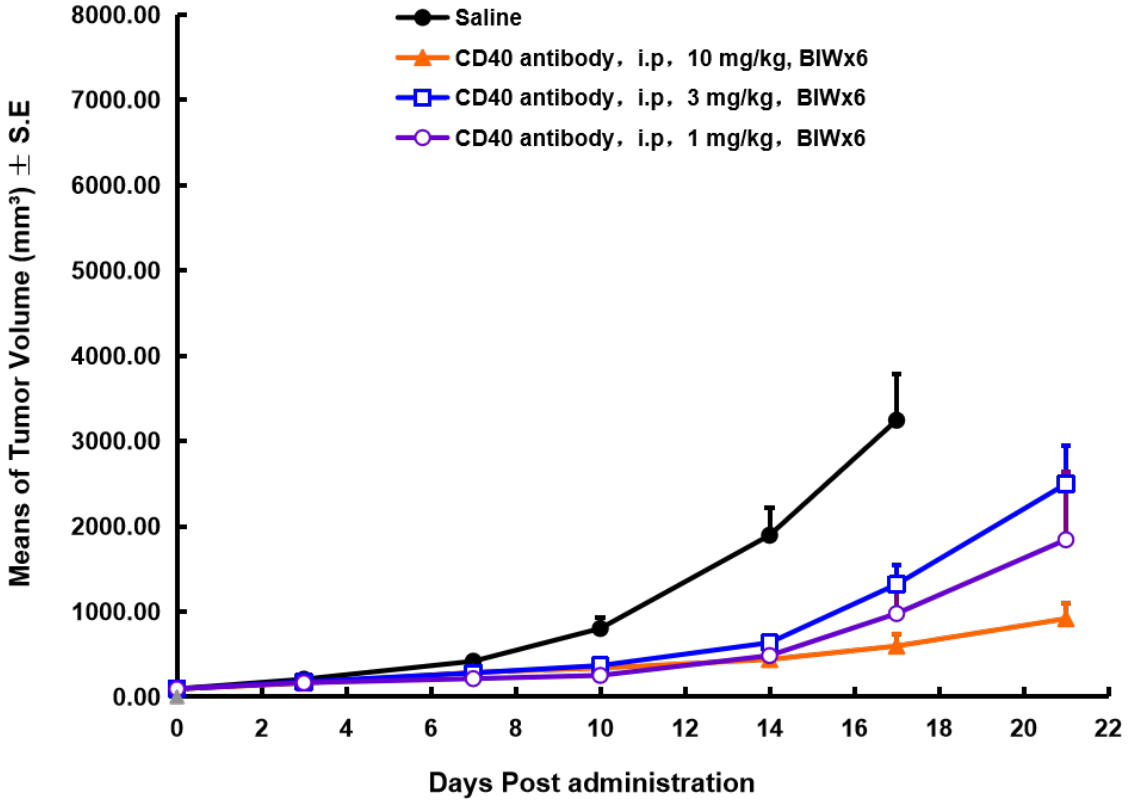



Figure 3. In vivo validation of anti-tumor efficacy in a MC38 tumor-bearing model of humanized CD40 mice. Humanized CD40 mice were inoculated with MC38 colon cancer cells. After the tumors grew to 100 mm3, the animals were randomly assigned into different group (n=8). The results indicated that the antibodies targeting human CD40 showed a very significant antitumor effect ($p<0.001$). Combination of anti-CD40 and anti-PD-1 is shown more significant anti-Tumor effect. (Cooperation with CrownBio)

**Case study 2：In vivo efficacy and safety evaluation of anti-human CD40 antibody using hCD40 mice**

I. In vivo anti-tumor effect of an anti-human CD40 antibody in hCD40 mice.


Saline
CD40 antibody，i.p，10 mg/kg, BIWx6
CD40 antibody，i.p，3 mg/kg，BIWx6
CD40 antibody，i.p，1 mg/kg，BIWx6
Means of Tumor Volume (mm³) ± S.E
8000.00
7000.00
6000.00
5000.00
4000.00
3000.00
2000.00
1000.00
0.00
0
2
4
6
8
10
12
14
16
18
20
22
Days Post administration

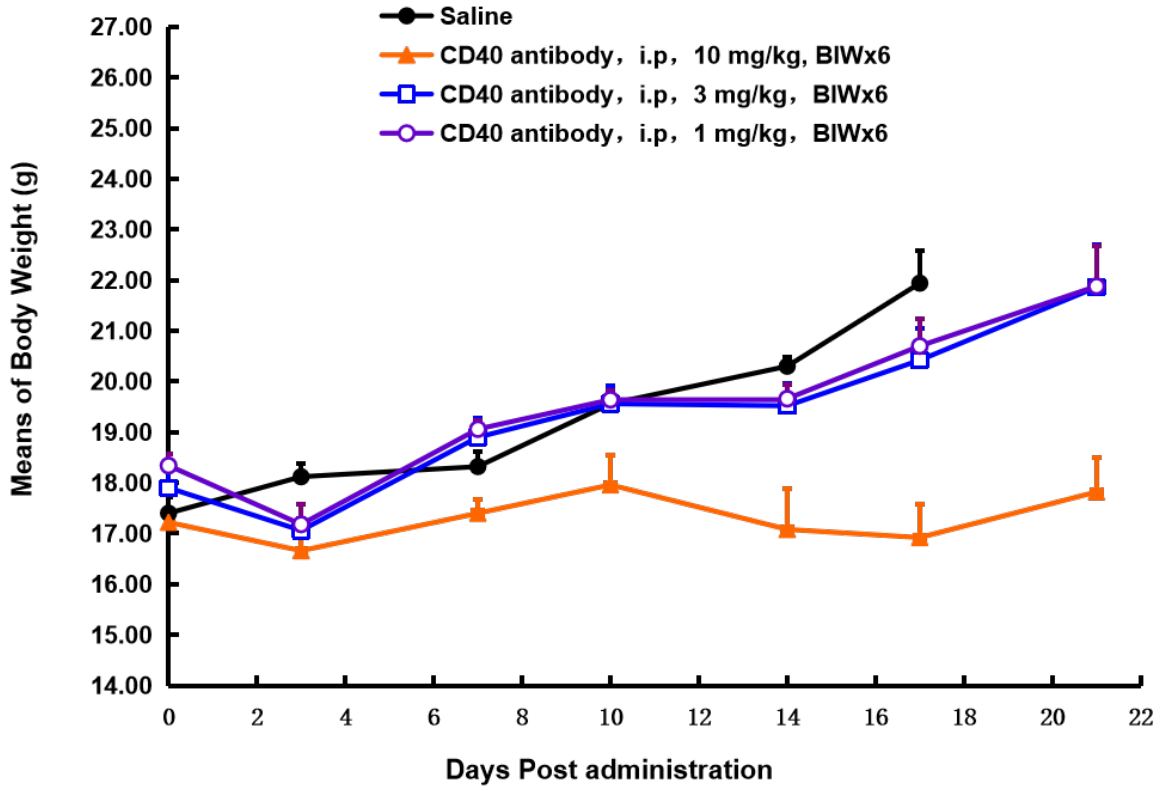
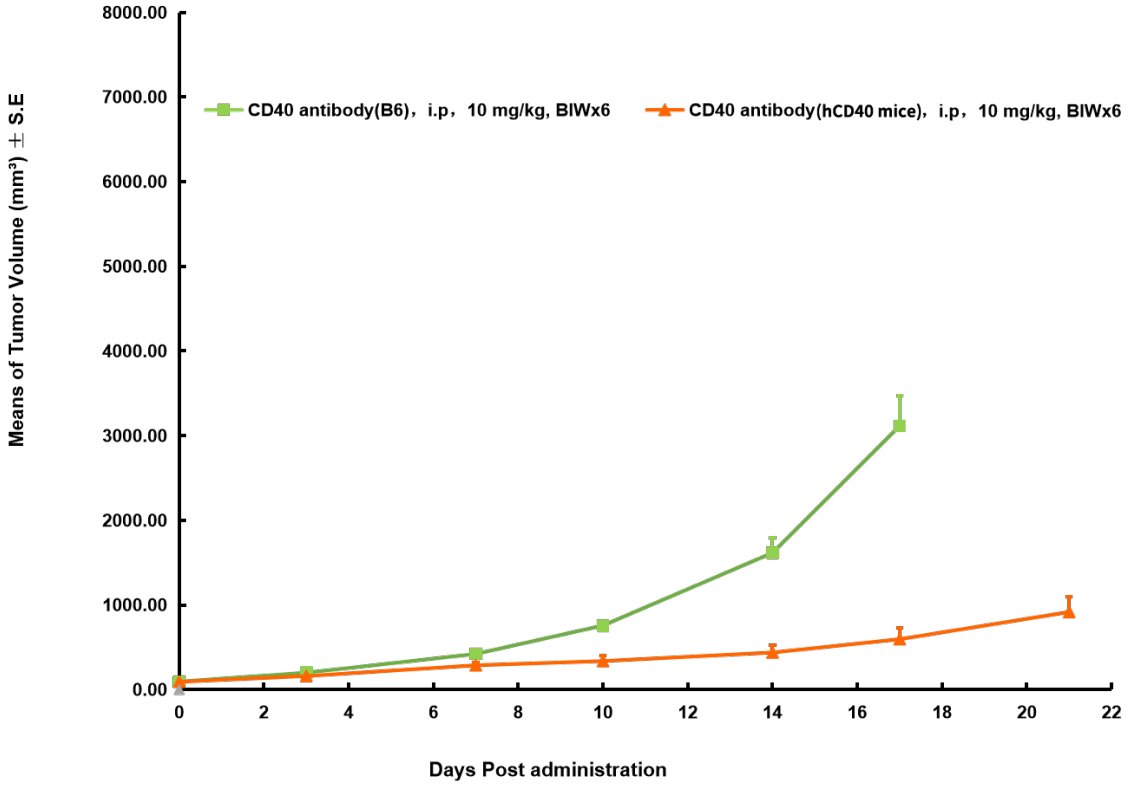

Saline
CD40 antibody，i.p，10 mg/kg, BIWx6
CD40 antibody，i.p，3 mg/kg，BIWx6
CD40 antibody，i.p，1 mg/kg，BIWx6
Means of Body Weight (g)
27.00
26.00
25.00
24.00
23.00
22.00
21.00
20.00
19.00
18.00
17.00
16.00
15.00
14.00
0
2
4
6
8
10
12
14
16
18
20
22
Days Post administration

Figure 4. In vivo validation of anti-tumor efficacy in a MC38 tumor-bearing model of humanized CD40 mice. Homozygous humanized CD40 mice were inoculated with MC38 colon cancer cells. The results showed that an anti-human CD40 antibody exerted a very significant anti-tumor effect , demonstrating that the humanized CD40 mice are a good in vivo model for validating the efficacy of antibodies targeting human CD40. Mean volume ± SEM of tumor tissues(A). Mean body weight ± SEM of mice(B).

II. Evaluation of anti-human CD40 toxicity in hCD40 mice.



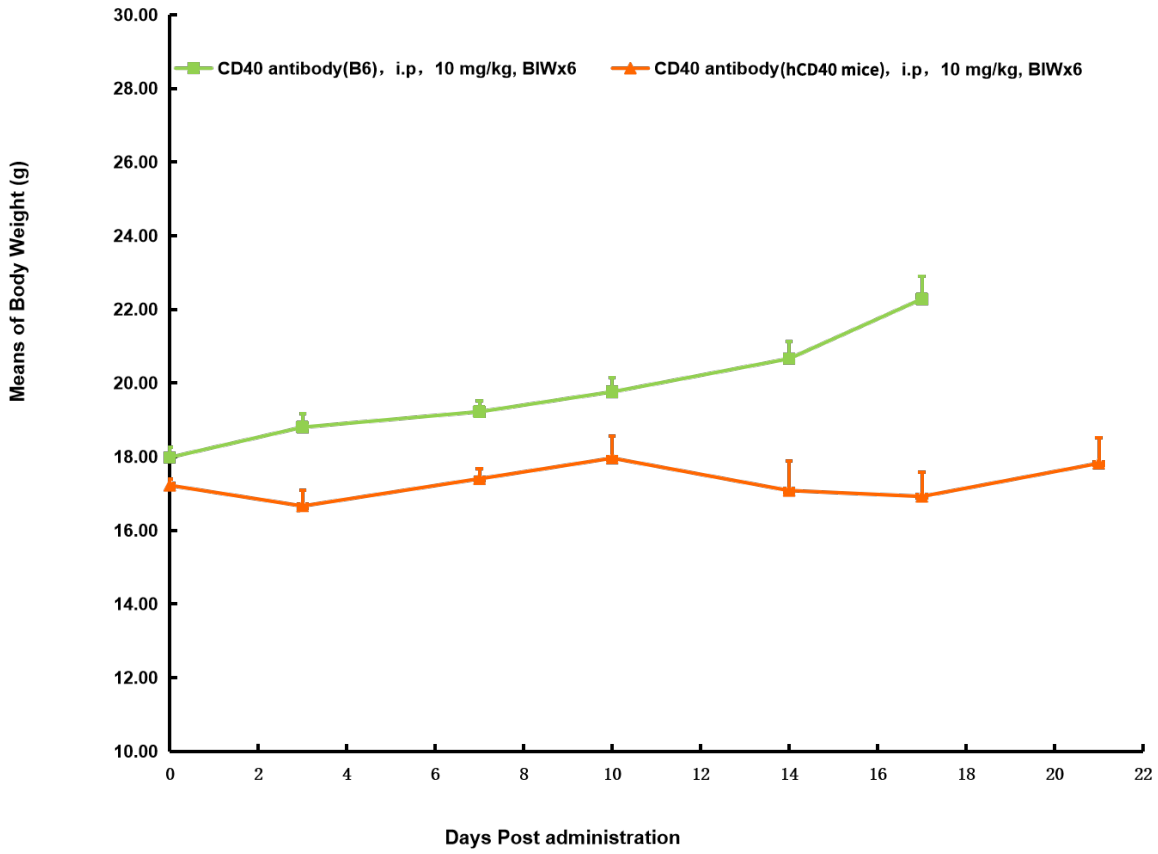
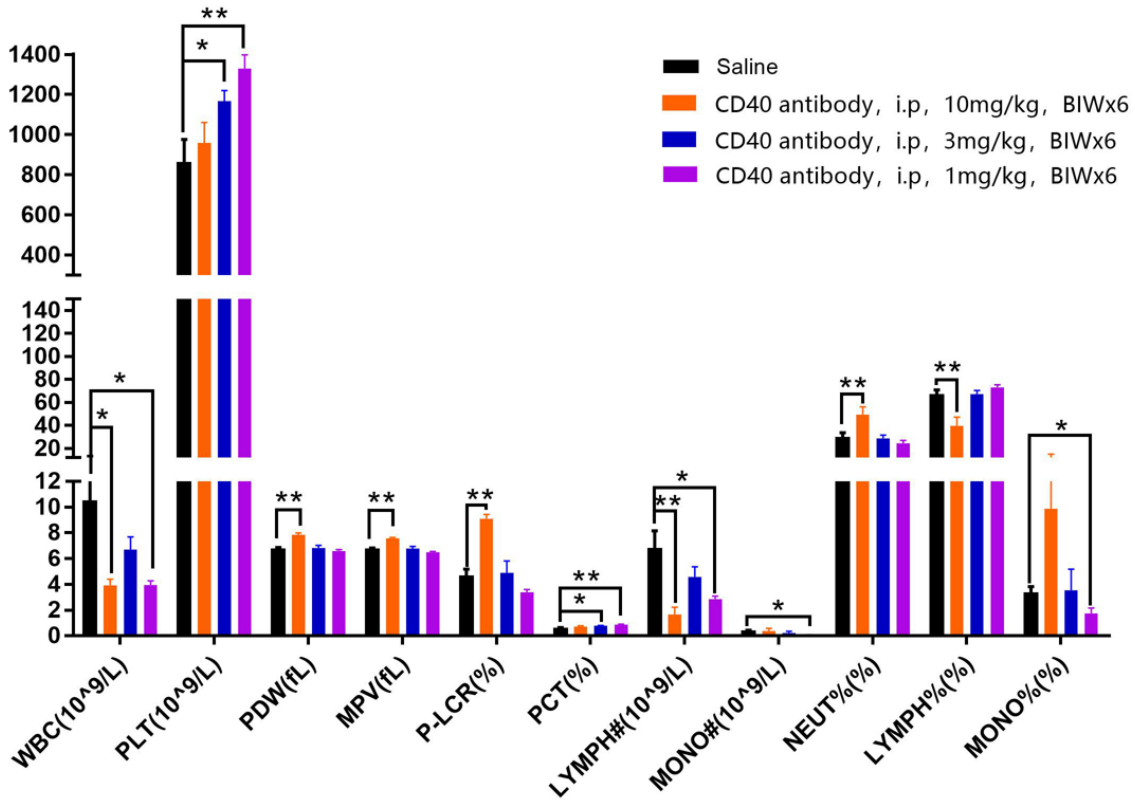
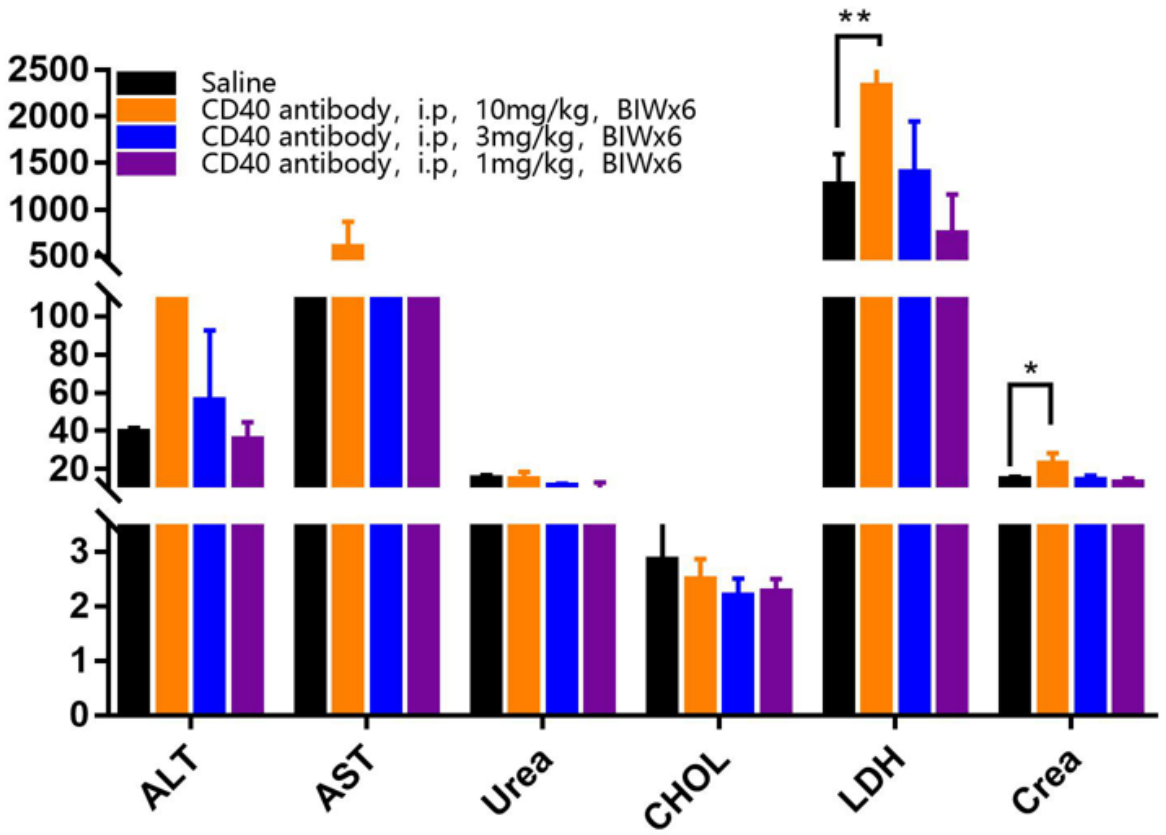



Figure 4. Tumor and weight changes over treated with anti-human CD40 antibody



| Abbreviation | Stand for |
|---|---|
| WBC | White Blood Cell Count |
| PDW | platelet distribution width |
| MPV | mean platelet volume |
| PLCR | platelet larger cell ratio |
| PLT | platelet |
| PCT | thrombocytocrit |
| LYMPH# | lymphocyte count |
| MONO# | monocyte count |
| NEUT% | neutrophil count |
| LYMPH% | lymphocyte ratio |
| MONO% | monocyte ratio |

Figure 5. Complete blood count (CBC) of anti-human CD40-antibody treated hCD40 mice




| Abbreviation | Stand for |
|---|---|
| ALT | Alanine transaminase |
| AST | Aspartate transaminase |
| Urea | Urea |
| CHOL | Cholesterol |
| LDH | lactate dehydrogenase |
| Crea | creatinine |

Figure 6. Blood chemistry of anti-human CD40-antibody treated hCD40 mice

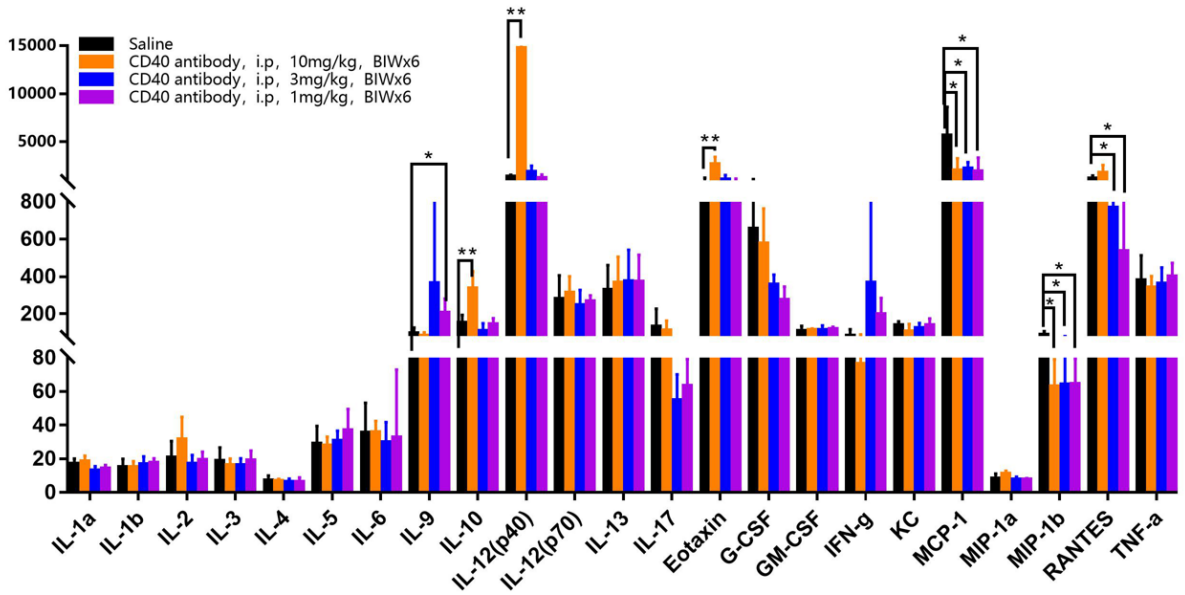



Figure 7. Cytokine analysis of anti-human CD40 antibody treated hCD40 mice.

Cytokine analysis of anti-human CD40 antibody treated MC38 tumor-bearing model of humanized CD40. Anti-human CD40 antibody treatment led to significant increase several cytokines including IL-12(p40), Eotaxin, etc.

# Immune Checkpoint Humanized Mouse Models

Being recognized as a top scientific breakthrough in 2013, cancer immunotherapy is predicted to be one of the most promising research areas for improving patient outcomes. Although many immunotherapy breakthroughs may still lie ahead, important clinical advances have been made in the past few years for some of the deadliest cancers, reaffirming the potential of immunotherapy for many types of patients.

However, it is worth noting that drug candidates developed to interfere with human proteins may not comparably interact with their murine counterparts. It is therefore critical to develop humanized mouse models to enable in vivo efficacy evaluation of cancer immunotherapies.

## Immune Checkpoint Humanized Mouse Models available at ingenious targeting laboratory

*4-1BB*
*CD40*
*CD47*
*CD73 (NT5E)*
*CTLA4 (C57BL/6)*
*CTLA4 (BALB/c)*
*KDR*
*LAG3*
*OX40*
*OX40/CTLA4*
*PD-1 (C57BL/6)*
*PD-1 (BALB/c)*
*PD-1/4-1BB*
*PD-1/CD40*
*PD-1/CTLA4*
*PD-1/LAG3*
*PD-1/OX40*
*PD-1/PD-L1*
*PD-1/TIGIT*
*PD-1/TIM3*
*PD-L1*
*PD-L1/CTLA4*
*PD-L1/LAG3*
*PD-L1/OX40*
*PD-L1/TIGIT*
*SIRPA*
*SIRPA/CD47*
*TIGIT*
*TIM3 (C57BL/6)*
*TIM3 (BALB/c)*
*TNFRSF1B*
***And more to come!***

To get to know more about these models, visit our website **www.genetargeting.com** or contact our scientific experts at **inquiry@genetargeting.com**

# About ingenious targeting laboratory

ingenious targeting laboratory (**ingenious**) has been a leading global provider of custom genetically modified mouse, rat, and rabbit models for over 20 years. As one of the very first mouse gene targeting companies, our trusted service is built on two decades' worth of successful animal model creation for investigators, organizations, and companies worldwide. Our models have been published in hundreds of journals including *Science*, *Nature*, and *Cell*, making us one of the most validated and respected production companies in the industry. We are excited to add catalog mouse models to our service repertoire by means of our collaboration with Shanghai Model Organisms Center (SMOC).

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